

Bring Music The Horizon: Music-Driven VR World Generation

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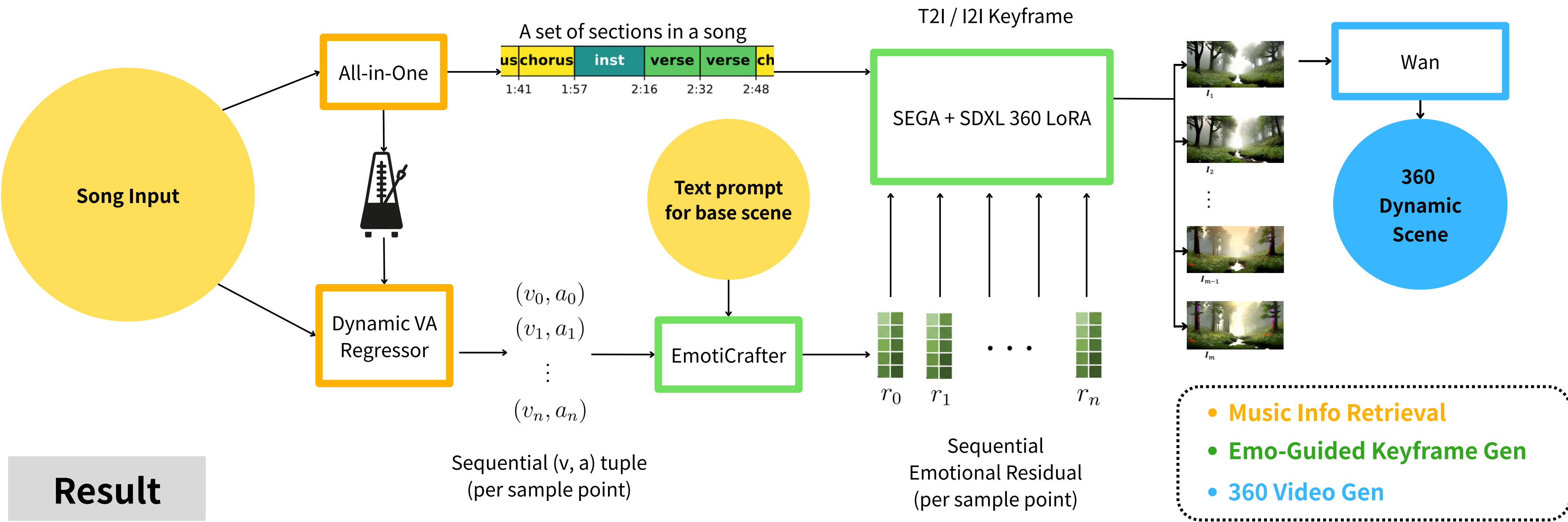
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Abstract

This research advances AI music visualization from single-perspective media to musically-aware VR experience by overcoming bottlenecks in **continuous emotion modeling** and **immersive scene presentation**. By integrating Music Information Retrieval, semantic embedding guidance techniques, and Diffusion models, we manage to synthesize a 360° immersive video that can respond to the emotional and acoustic dynamics of the given song. This project establishes a solid foundation for immersive content and future transmedia creation.

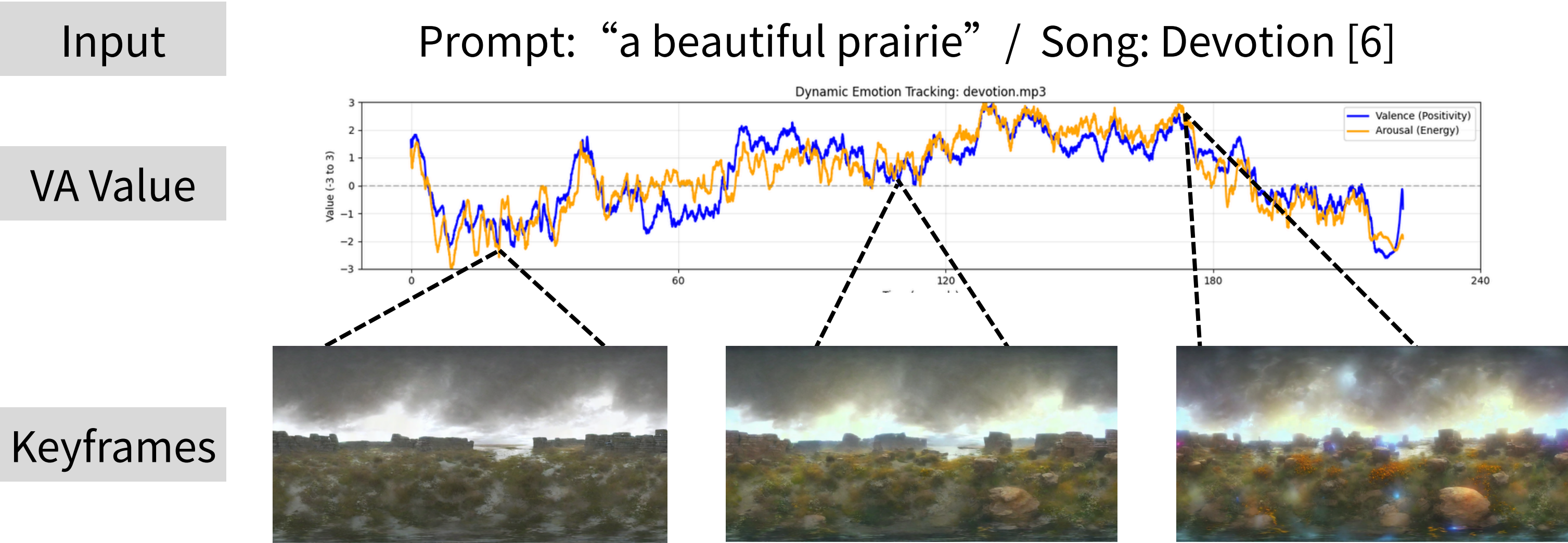
Method

Our pipeline consists of a Music Information Retrieval module, an Emotion-guided Keyframe Generation module, and a 360° video generation module. First, **All-in-one[1]** will extract the timestamp of each downbeat. Then, our **VA[2] Regressor** will extract the implied emotion, also known as the **VA value[2]**, at each downbeat. Next, **EmotiCrafter[3]** will generate the "emotional residual" based on the VA value, which is then used with **SEGA[4]** **guidance term** to guide the diffusion process of the keyframes. Lastly, we will use **Wan-I2V[5]** to generate 3 bars of dynamic scene for each keyframe, and then **Wan-flf2v[5]** to generate the transition video between each dynamic scene. By concatenating the dynamic scenes and their transitions, we can obtain the immersive visualization of the song in the form of 360° video.



Result

The pipeline generates 360° video that can be played in a VR headset to get immersive music visualization experience. Each keyframe will align with the emotion of their corresponding timepoints in the song.



Reference

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[5] Team Wan et al., "Wan: Open and Advanced Large-Scale Video Generative Models," arXiv:2503.20314, 2025.
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Demo

